

5(a)	<i>ignore single letters for the bases</i> A <i>purines</i> = adenine <u>and</u> guanine ; <i>in either order</i> <i>pyrimidines</i> = cytosine <u>and</u> uracil ; <i>in either order</i> B phosphodiester (bond) ; C ribose ;	4
5(b)	any three from: <i>penalise once if incorrect, pathogen / type of organism</i> aerosol infection / droplet infection / ref. to airborne droplets ; <i>in context of droplets containing, pathogen / virus</i> infected person, exhales / expires / breathes out / coughs / sneezes / AW, (airborne droplets) ; (airborne droplets) breathed in / AW, by, uninfected / healthy, person; <i>idea of</i> transmitted when uninfected person touches an infected surface and puts fingers into, mouth / nose ; R contact <i>without qualification</i>	3

Question	Answer	Marks
5(c)(i)	trachea bronchus / bronchi bronchiole / bronchioles ; A types of bronchiole	1
5(c)(ii)	A CO for carbon monoxide A Hb / hb, for haemoglobin any two from 1 forms carboxyhaemoglobin ; A binds to haemoglobin (permanently) 2 less haemoglobin available to bind oxygen / haemoglobin has higher <u>affinity</u> for carbon monoxide (than for oxygen) ; 3 reduces, percentage saturation of haemoglobin (with oxygen) / AW ; A less, oxygen binds to haemoglobin / oxyhaemoglobin forms I prevents oxygen binding <i>if mp2 and mp3 not gained allow one mark for statements such as reduces capacity for binding of oxygen to <u>hb</u> / less oxygen carried by <u>hb</u> / decreases the oxygen carrying capacity of <u>hb</u></i> 4 decreased quantity of oxygen transported / AW ; 5 damages, endothelium / tunica intima / lining of blood vessels ;	2
5(d)(i)	(steady) increase in number of (estimated) deaths (from measles, over time / from 2000 to 2017) ; A (relatively) constant then (steady) increase in number of (estimated deaths over time)	1

Question	Answer	Marks
5(d)(ii)	any three from: I ref. to cost or mutations 1 as (global) population increased, higher, number / proportion, of unvaccinated ; ora 2 <i>ref. to</i> HIV link to measles ; e.g. increase in number of people living with HIV increases spread of measles increase in cases of HIV / AIDS, measles can be an opportunistic disease R spread of HIV / AIDS weakened immune system in people with HIV / AIDS so measles more likely to occur (in unvaccinated) 3 less / no, herd immunity (so increased chance of getting measles) ; A <i>ref. to</i> larger, number / proportion, unvaccinated will increase risk of spreading disease <i>context is community</i> 4 one example of a difference between unvaccinated versus vaccinated ; e.g. risk of <u>death</u> from measles is higher in unvaccinated unvaccinated do not have, (artificial active) immunity / memory cells <i>context is individuals</i> 5 suggestion as to why fewer people are vaccinated ; e.g. reluctance, of parents to get child vaccinated reluctance because of, cultural / religious, beliefs difficulty in obtaining vaccine <i>in context of remote communities</i> collapse of vaccination programmes in some countries / AW lack of, trained professionals / health facilities, to deliver vaccine lack of education about benefits of vaccine / AW 6 suggestion as to why measles cases (and so deaths) may increase ; e.g. (greater proportion of people) malnourished / lack vitamin A (greater proportion of people) living in, overcrowded / poor, conditions I overcrowded countries measles is easily spread (with increase in population) A very infectious measles is more easily spread in overcrowded conditions	3

Question	Answer	Marks
5(d)(ii)	7 example of spread (of measles) owing to movement of people between countries ; e.g. (unvaccinated) people with measles arriving to a country and spreading disease unvaccinated people travelling to countries with the disease, catching measles and taking it back home people arriving with measles to countries with low vaccination rates 8 AVP ; e.g. ref. to, no / any specific difficulty for, contact tracing lack of, isolation / quarantining poor epidemiological records so authorities cannot react to outbreaks improved, recording / reporting, of deaths over the time period	
5(e)	any two from: (high count owing to), cancerous / uncontrolled mitosis of, white blood cells ; A malignant cells A uncontrolled division white blood cells (in child with leukaemia) cannot, function (in immunity) / protect / example of function ; A undifferentiated / unspecialised (so) weakened immune system (and more likely of developing measles) ; AW vaccination is, less effective / not effective / gives weak immune response ; secondary immune response, does not occur / too slow (after vaccination) ; AVP ; has memory cells but clonal expansion, does not occur / is faulty	2